

Mehmet KORUTURK

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SUMMARY

I am a Ph.D. student in Electrical Engineering at Texas A&M University, with an M.S. in Electrical Engineering from Virginia Tech and research experience in reinforcement learning, optimization, and grid-interactive power systems. I am interested in developing energy-management and decision-making systems that are efficient, safe, interpretable, robust, and capable of continually adapting to changing grid conditions, uncertainty, and disturbances, enabling learning-based control to be trusted in real, mission-critical energy infrastructure.

EDUCATION

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| August 2026 – Present | Ph.D. in Electrical Engineering
Texas A&M University (TAMU)
Advisor: Robert Balog |
| August 2024 – May 2026 | M.S. in Electrical Engineering
Virginia Polytechnic Institute and State University (Virginia Tech),
GPA: 3.96/4.00
Advisor: Ming Jin
Thesis: Reinforcement Learning Benchmarking for Sustainable Energy Systems: Perturbation Robustness, Safety Constraints, and Multi-Agent Coordination
Relevant Coursework: <ul style="list-style-type: none">• Power System Planning (Prof. Lamine M. Mili)• Advanced Alternative Energy Systems (Prof. Saifur Rahman)• Advanced Microgrids (Prof. Ali Mehrizi-Sani)• Computational Methods in Power (Prof. Lamine M. Mili) |
| August 2017 – May 2022 | B.Sc. in Electrical Engineering
Yildiz Technical University (YTU), GPA: 3.21/4.00
Advisors: Prof. Ozan Erdinc , Prof. Ali Rifat Boynuegri |

RESEARCH EXPERIENCE

Graduate Research Assistant, Texas A&M University	Aug 2026 – Present
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Graduate Research Assistant, Virginia Tech	Aug 2024 – May 2026
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- Built RL benchmarks for power-system control under uncertainty.
- Developed RL controllers for PV, ESS, EV, and FCEV systems.
- Designed resilience-aware optimization for post-disaster energy management.
- Evaluated robustness and stability of learning-based policies.

PUBLICATIONS

- Koruturk, Mehmet**, Alper Kagan Candan, et al. (July 2025). “Post-Disaster Energy Management for Homes Using Limited-Capacity Fuel Cell Electric Vehicle Batteries”. In: *Renewable Energy*. Under review.
- Koruturk, Mehmet**, Artun Sel, et al. (June 2025). “Optimization of End-Consumer Energy Management with PV, ESS, EV, and Peer-to-Peer Trading in Smart Grids”. In: *IEEE INTCEC 2025*. URL: <https://ieeexplore.ieee.org/document/11255815>.
- Sel, Artun, Will Barrett, and **Mehmet Koruturk** (June 2025). “Estimation of Regions of Attraction for Nonlinear Systems via Coordinate-Transformed TS Models”. In: *IEEE INTCEC 2025*. URL: <https://ieeexplore.ieee.org/document/11255827>.
- Sel, Artun, **Mehmet Koruturk**, et al. (July 2025). “Estimation of Regions of Attraction for Nonlinear Systems via Coordinate-Transformed TS Models and Piecewise Quadratic Lyapunov Functions”. In: *IEEE INTCEC 2025*. URL: <https://ieeexplore.ieee.org/document/11256030>.
- Koruturk, Mehmet** et al. (2026). “Post-Disaster Home Energy Management with Fuel Cell Vehicles: Adaptive MILP+RL Coordination via Deviation-Based Mode Switching Under Islanding Conditions”. In: *Applied Energy*. In preparation.
- Koruturk, Mehmet**, Alper Kagan Candan, Ali Rifat Boynuegri, et al. (2026). “Post-Disaster Home Energy Management with Fuel Cell Vehicles: Evaluating Reinforcement Learning Against MILP Under Islanding Conditions”. In: *2026 8th Global Power, Energy and Communication Conference (GPECOM)*, pp. 1058–1063. DOI: [10.1109/GPECOM70462.2026.11578574](https://doi.org/10.1109/GPECOM70462.2026.11578574). URL: <https://doi.org/10.1109/GPECOM70462.2026.11578574>.
- Koruturk, Mehmet**, Bilgehan Sel, et al. (2026). “SustainRL-Bench: A Three-Axis RL Benchmark for Sustainable Energy Systems”. In: *IEEE SmartGridComm 2026*. Accepted.
- Yigit, Hayri, Ali Toprak, and **Mehmet Koruturk** (2026). “Hybrid Battery–Wireless Power Transfer System for Range-Extended Disaster-Relief UAVs: A Coupled Thermal–Electrical Battery Sizing Optimization”. In: *Energies*. Accepted.

INDUSTRY EXPERIENCE

- Electrical Engineer, Huawei Technologies Co. Ltd.** Jul 2022 – Jul 2023, Istanbul, Turkey
- Led technical deployment and validation of 4G/5G base station installations.
 - Conducted system-level acceptance testing and performance verification.
 - Coordinated site implementation for large-scale telecom infrastructure projects.

PROFESSIONAL SERVICE

- Peer Reviewer,
IEEE Transactions on Automation Science and Engineering (T-ASE) (2025–Present)
- Core Organizing Committee Member,
Power and Energy Center (PEC) 2026 Annual Conference, Virginia Tech

AWARDS

- Awarded *Turkey’s Ministry of Education Scholarship for Graduate Studies in USA*
(2022–Present)

SKILLS

Programming	Python, MATLAB
Engineering Tools	PSCAD, AutoCAD
Productivity	MS Office